

```

34. # Sharpe Ratio (assuming 0% risk-free rate for simplicity)
35. sharpe_ratio = annualized_return / volatility if volatility > 0 else 0
36.
37. # Maximum Drawdown
38. cumulative_returns = (1 + returns).cumprod()
39. peak = cumulative_returns.expanding().max()
40. drawdown = (cumulative_returns / peak) - 1
41. max_drawdown = drawdown.min()
42.
43. # Calmar Ratio (annualized return divided by maximum drawdown)
44. calmar_ratio = annualized_return / abs(max_drawdown) if max_drawdown < 0 else 0
45.
46. # Win rate
47. trades = backtest_results[backtest_results['Trade'] != 0]
48. wins = trades[trades['Portfolio Value'] > trades['Portfolio Value'].shift(1)]
49. win_rate = len(wins) / len(trades) if len(trades) > 0 else 0
50.
51. # Average profit/loss per trade
52. if len(trades) > 0:
53.     trades['Trade PnL'] = trades['Portfolio Value'].diff()
54.     avg_profit = trades[trades['Trade PnL'] > 0]['Trade PnL'].mean() if len(trades[trades['Trade PnL'] > 0]) > 0 else 0
55.     avg_loss = trades[trades['Trade PnL'] < 0]['Trade PnL'].mean() if len(trades[trades['Trade PnL'] < 0]) > 0 else 0
56.     profit_loss_ratio = abs(avg_profit / avg_loss) if avg_loss < 0 else 0
57. else:
58.     avg_profit = 0
59.     avg_loss = 0
60.     profit_loss_ratio = 0
61.

```